FOCS_StudF03: Student's Progress Report

Tunku Abdul Rahman University of Management and Technology
Faculty of Computing and Information Technology
Industrial Training Progress Report

Activity Log



Name of Trainee:	Yeap Zi Jia
Name of Company:	SOPHIC AUTOMATION SDN BHD.
Month/Year	JUNE/2026
	(E.g. JAN/2022)

Week	Projects / Activities
1	**Description of project/activity** **Activity 1: Excel Import – Draft Pool** Developed and implemented the Excel import functionality for the Draft Pool module, enabling users to upload and process draft pool data efficiently into the RTM system. Implemented validation and data mapping logic to ensure uploaded Excel data matched the required system structure and business rules. **Activity 2: Excel Export – ECN Management** Developed the Excel export feature for the ECN Management module, allowing users to generate and download ECN-related data in Excel format for reporting and analysis purposes. Enhanced data retrieval and formatting processes to ensure accurate and reliable report generation. **Activity 3: Excel Import – Custom Data Sync** Implemented enhancements to the Excel import process to support synchronization of custom data fields not available in the RTM database, including Golden SVID, Control Type, Chart Type, and Fixed Control Limit. Developed custom handling logic to process and preserve imported information during synchronization.

Activity 4: Services – Update Master DB When ECN Status Changes

Developed backend service logic to automatically update the Master Database whenever an ECN status change occurs. Enhanced the synchronization workflow to maintain data consistency across related modules and ensure updates are triggered based on business process events.

A total of Time spent on the activity

- Activity 1: Excel Import – Draft Pool is 11 hours
- Activity 2: Excel Export – ECN Management is 11 hours
- Activity 3: Excel Import – Custom Data Sync is 11 hours
- Activity 4: Services – Update Master DB When ECN Status Changes is 12 hours

Technology/application used to complete the activity

- ASP.NET Web API (C#)
- Visual Studio
- RTM System
- SQL Database
- Excel File Processing
- Entity Framework
- Swagger API Testing
- VDI Environment

What challenges have you faced (if any)?

Faced challenges in implementing the Draft Pool Excel import functionality, as uploaded files required proper validation and field mapping to ensure the data matched the expected system structure and could be processed successfully without errors.

Additionally, developing the ECN Management Excel export feature required accurate data retrieval, transformation, and formatting to ensure generated reports were complete, consistent, and suitable for business reporting purposes.

	Another challenge involved synchronizing custom data fields that were not directly supported by the RTM database schema. Additional processing and mapping logic were required to preserve and handle fields such as Golden SVID, Control Type, Chart Type, and Fixed Control Limit during the import process. Ensuring the Master Database was updated correctly when ECN statuses changed also required careful handling of backend service workflows to maintain data consistency and prevent synchronization issues across related modules. **What skills/knowledge have you learned from the activities?** Gained practical experience in developing Excel import and export functionalities within the RTM system, including file processing, validation, data mapping, and report generation workflows. Improved understanding of handling complex data synchronization requirements, particularly when integrating custom fields that differ from existing database structures and schemas. Enhanced backend development skills by implementing automated service logic that updates system data based on business workflow events and status changes. Also strengthened testing, debugging, and troubleshooting abilities while validating data synchronization processes, export functionality, and database update operations to ensure system reliability and data accuracy.
2	**Description of project/activity** **Activity 1: K2 Integration – Excel Import & Template Download** Integrated the Excel import functionality and Excel template download feature into the K2 system, enabling users to import data and download predefined templates directly from the application. Performed integration and

validation testing to ensure compatibility with existing K2 workflows and system configurations.

Activity 2: K2 Integration – ECN Management Excel Export

Integrated the ECN Management Excel export functionality into the K2 system, allowing users to generate and download ECN-related data in Excel format for reporting and analysis purposes. Verified data accuracy and export consistency to ensure reliable report generation.

Activity 3: Services – Update Master DB When ECN Status Changes

Continued the development of an executable (EXE) application designed to automatically update the Master Database whenever ECN status changes occur. Implemented synchronization logic and investigated scheduled execution mechanisms to support automated background processing and data updates.

A total of Time spent on the activity

- Activity 1: K2 Integration – Excel Import & Template Download is 15 hours
- Activity 2: K2 Integration – ECN Management Excel Export is 17 hours
- Activity 3: Services – Update Master DB When ECN Status Changes is 13 hours

Technology/application used to complete the activity

- ASP.NET Web API (C#)
- Nintex K2
- Visual Studio
- SQL Database
- Excel File Processing
- Windows Task Scheduler
- Swagger API Testing
- VDI Environment

What challenges have you faced (if any)?

Faced challenges while integrating the Excel import and template download functionalities into the K2 environment, as the implementation needed to remain compatible with existing workflows, user processes, and system configurations. Additional testing and configuration adjustments were required to ensure smooth functionality within the integrated environment.

Another challenge involved validating the ECN Management Excel export feature to ensure the generated reports accurately reflected the source data. Careful verification and testing were performed to confirm that exported files contained complete and consistent information for reporting purposes.

Developing the EXE application for automatic Master Database updates also required understanding scheduled execution mechanisms and ensuring reliable synchronization whenever ECN status changes occurred. Additional testing was necessary to verify that background execution processes operated correctly and triggered updates as expected.

What skills/knowledge have you learned from the activities?

Gained practical experience in integrating existing system functionalities into the K2 environment, including Excel import, template download, and export features. Improved understanding of system integration processes, compatibility verification, and user workflow considerations.

Enhanced knowledge of Excel file processing, report generation, template management, and export validation within enterprise applications. Also developed stronger testing and troubleshooting skills while validating integrated functionality across multiple systems.

Improved backend development knowledge through the development of an executable application for automated processing. Learned more about scheduled task execution, automation concepts, and data synchronization workflows used to maintain consistency across enterprise systems.

	Also strengthened problem-solving abilities by addressing integration challenges, validating system behavior, and ensuring reliable execution of automated database update processes.
3	**Description of project/activity** **Activity 1: Services – Update Master DB When ECN Status Changes** Completed the development and testing of an executable (EXE) application that automatically updates the Master Database whenever ECN status changes occur. The solution supports scheduled execution and automated synchronization between systems to ensure data consistency and minimize manual intervention. **Activity 2: FDC API Data Mapping Study** Conducted analysis and study of the FDC API data structure and mapping requirements to understand data relationships, field mappings, and integration requirements. Reviewed API responses and business requirements to identify how source data should be transformed and synchronized into the target system for future development activities. **Activity 3: Custom Data Sync Excel Function – UI Development** Developed the user interface for the Custom Data Sync Excel function, enabling users to manage Excel-based data synchronization processes. Designed UI components and validation mechanisms to improve usability and prepare the interface for future backend integration and synchronization functionality. **A total of Time spent on the activity** • Activity 1: Services that update Master DB When ECN Status Changes is 8 hours • Activity 2: FDC API Data Mapping Study is 10 hours • Activity 3: Custom Data Sync Excel Function that UI Development is 12 hours

Technology/application used to complete the activity

- ASP.NET Web API (C#)
- Visual Studio
- SQL Database
- Windows Task Scheduler
- REST API
- FDC API
- VDI Environment

What challenges have you faced (if any)?

Faced challenges in ensuring the EXE application reliably updated the Master Database whenever ECN status changes occurred. Additional testing was required to verify scheduled execution behavior, synchronization accuracy, and overall data consistency across related systems.

Another challenge involved understanding the FDC API structure and identifying accurate data mappings between source and target systems. The API responses contained multiple related entities and data relationships, requiring detailed analysis of field definitions, business requirements, and integration expectations before implementation could proceed.

Developing the Custom Data Sync Excel function user interface also required careful consideration of usability, input validation, and future backend integration requirements. The interface needed to support efficient user interaction while remaining flexible enough to accommodate future synchronization enhancements.

What skills/knowledge have you learned from the activities?

Gained practical experience in developing and testing automated synchronization services, including scheduled background execution and database update processes. Improved understanding of synchronization workflows and strategies for maintaining data consistency across multiple systems.

	Enhanced knowledge of API integration concepts through the analysis of FDC API structures, request and response data, and field mapping requirements. Learned how to interpret business requirements and translate them into technical integration specifications. Improved front-end development skills by designing and implementing user interface components for Excel-based data synchronization processes. Also strengthened problem-solving, testing, and troubleshooting abilities while validating synchronization workflows and preparing integration-related features for future development.
4	**Description of project/activity** **Activity 1: RTM Data Mapping Analysis & SQL Query Development** Conducted analysis of the RTM database structure and data mapping requirements to understand the relationships between related tables. Developed SQL queries to retrieve the required data from RTM tables, ensuring accurate data extraction to support subsequent integration and mapping activities. **Activity 2: FDC Swagger Login Issue Troubleshooting** Investigated the login issue encountered when accessing the FDC Swagger API and collaborated with the client's IT team to identify the root cause. Coordinated troubleshooting activities, verified authentication and system configurations, and confirmed the implemented solution successfully restored API accessibility. **Activity 3: FDC Swagger API Testing & Data Mapping Verification** Performed testing on the FDC Swagger API to retrieve data and verify API responses against the required data mapping specifications. Reviewed returned payloads and validated field mappings to ensure the retrieved data aligned with integration requirements and supported accurate data synchronization. **A total of Time spent on the activity**

- Activity 1: RTM Data Mapping Analysis & SQL Query Development is 8 hours
- Activity 2: FDC Swagger Login Issue Troubleshooting is 10 hours
- Activity 3: FDC Swagger API Testing & Data Mapping Verification is 12 hours

Technology/application used to complete the activity

- Microsoft SQL Server
- SQL Server Management Studio (SSMS)
- FDC Swagger API
- REST API
- Visual Studio
- Postman / Swagger
- RTM Database
- VDI Environment

What challenges have you faced (if any)?

Faced challenges in understanding the relationships between multiple RTM database tables and determining the appropriate SQL queries required to retrieve the necessary data for integration and mapping purposes. Careful analysis of the database schema and data relationships was required to ensure the extracted information was accurate and complete.

Another challenge involved resolving the FDC Swagger login issue, which required close collaboration with the client's IT team to investigate authentication failures and system access configurations. Multiple troubleshooting sessions and verification activities were conducted before API accessibility was successfully restored.

Additionally, validating the FDC API responses against the required data mapping specifications required detailed analysis of the returned payloads to ensure field mappings, data relationships, and business requirements were correctly aligned. Thorough testing was performed to verify the accuracy and

	consistency of the retrieved data before proceeding with further integration work. **What skills/knowledge have you learned from the activities?** Gained practical experience in analyzing enterprise database structures and developing SQL queries to retrieve data from multiple related RTM tables. Improved understanding of database relationships, data extraction techniques, and data mapping processes used in system integration projects. Enhanced troubleshooting and communication skills by collaborating with the client's IT team to resolve API authentication and connectivity issues. Learned the importance of cross-team coordination when diagnosing and resolving infrastructure-related problems. Strengthened knowledge of REST API testing, response validation, and data mapping verification through testing FDC Swagger API endpoints and analyzing returned payloads. Also improved analytical and problem-solving skills by ensuring consistency between source system data, database structures, and integration requirements.
5	**Description of project/activity** **Activity 1: Fake API Orchestration & Data Aggregation** Development Developed and orchestrated a backend integration endpoint (fdc-fakewithreadconfig-summary) by fetching data across multiple chained DreamFactory API endpoints (Tools, Collections, and Limits). Implemented dynamic JSON data mapping and developed a parallel array extraction algorithm to pivot columnar limit arrays into a unified, frontend-friendly response object. **A total of Time spent on the activity** • Activity 1: Fake API Orchestration & Data Aggregation Development is 18 hours

Technology/application used to complete the activity

- Microsoft SQL Server
- SQL Server Management Studio (SSMS)
- FDC Swagger API
- REST API
- Visual Studio
- Postman / Swagger
- RTM Database
- VDI Environment

What challenges have you faced (if any)?

Faced challenges in determining the correct mapping hierarchy when fetching limits data, as it required extracting the toolId dynamically from nested chambers arrays rather than the root tool object. Another significant challenge was processing the Limits API response, which returned data in parallel columnar arrays instead of individual objects. This required designing a custom JSON pivoting algorithm to accurately map and flatten the arrays into structured Data Transfer Objects (DTOs) so the frontend could consume them easily. Careful URL encoding of query parameters was also required to prevent missing collection reference errors during the chained API calls.

What skills/knowledge have you learned from the activities?

Gained deep practical experience in aggregating, orchestrating, and parsing complex dynamic JSON objects in C# using Newtonsoft.Json. Specifically improved skills in transforming and pivoting parallel data structures into standard entity models for API consumers. Enhanced problem-solving abilities by successfully designing nested iteration logic to chain multiple REST API calls (Tools -> Collections -> Limits) sequentially. Strengthened understanding of how to debug API request formulations and manage dynamic query parameters effectively in backend proxy endpoints.

Suggestions / Comments / Additional information (if any):

Leave Application / Leave Taken

1.From (dd/mm/yyyy): N/A to (dd/mm/yyyy) N/A (N/A)

2. Reasons for taking leave: N/A

3. Total number of days taken: N/A

I hereby declare that the information given above is correct.

Signature: *Yeap* **Date** (dd/mm/yyyy): 30/06/2026

Endorsement by the Company Supervisor:

The above is a true record of activities taken by the trainee in the captioned week.

Signature of Supervisor:	
Name of Supervisor:	Teoh Wei De
Date *(dd/mm/yyyy)*:	30/06/2026
Email:	weide.teoh@sophicautomation.com
Mobile / Office Contact No.:	019 - 466 1839

Company Stamp:

